

Qiaoyuan Zheng

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💻 qiaoyuan-zheng.com | 📍 Zurich, Switzerland | 🎓 MSc in Data Science



MSc student in Data Science at ETH Zürich, researching reliable LLM evaluation, interpretable capability assessment, and privacy in language agents.

Education

ETH zürich **ETH Zürich**

Zurich, Switzerland

M.Sc. DATA SCIENCE

2025 - Present

- GPA: 5/6
- Course: Probabilistic AI | Large Language Models | Big Data | Computational Statistics | Large-Scale AI Engineering
- Special Project-oriented course: AI Center Project | Research in DS | Data Science Lab

S **Sorbonne University**

Paris, France

B.Sc. COMPUTER SCIENCE/MATHEMATICS

2022 - 2025

- Math Course: Linear Algebra | Digital and Functional Series | Topology | Theory of Measurement | Theory of Probability
- CS Course: Algorithms | Data Structure | Object-oriented Programming | Functional Programming | Advanced C
- GPA: 17.10/20, 3.96/4.0 (WES), L3 Mention: Bien

Research Experience

ETH AI CENTER **ETH AI Center**

Zurich, Switzerland

RESEARCH ASSISTANT

2026 - Present

- Develop a multilingual benchmark for evaluating legal compliance in large language models
- Building reproducible pipelines for scenario generation, independent review, and model evaluation
- Collaborate with legal experts to assess scenario plausibility and legal validity
- Investigate benchmark applications for LLM post-training

Apertus LLM Evaluation Python

ESIEE **ESIEE Paris**

Paris, France

2023 - 2023

RESEARCH ASSISTANT · GITHUB

- Implemented Transformer-based models and PatchTST for time-series forecasting under the supervision of Dr. You Jiang
- Adapted data and training pipelines, comparing model performance and computational efficiency against U-Net 24 on synthetic datasets

Python PyTorch Time-Series Forecasting

Publications

POLAR-Bench: A Diagnostic Benchmark for Privacy-Utility Trade-offs in LLM Agents

2026

Qiaoyuan Zheng, Yiqu Yang, Qi Gao, Imanol Schlag

NeurIPS 2026 E&D Track

- Introduced a diagnostic benchmark covering 7,852 samples across 10 domains to evaluate how LLM agents balance task utility with user-defined privacy constraints.
- Separated privacy-policy dimensions and adversarial strategies in a 5×5 evaluation design, using deterministic scoring to identify specific failure conditions.
- Identified substantial differences in privacy protection between frontier models and smaller open-weight models, with the weakest evaluated models leaking over half of protected attributes.

Are Near-Tied LLM Rankings Robust to Family-DIF-Guided Benchmark Recomposition?

2026

Qiaoyuan Zheng, Yiqu Yang

NeurIPS TAE Workshop, Under review at ICLR 2027

- Investigated the robustness of near-tied LLM rankings across five benchmarks using a spectral approximation to multidimensional item response theory and differential item functioning analysis.
- Used owner-disjoint evaluation folds and matched-random controls to distinguish systematic effects of benchmark composition from generic subtest variation.

What Makes Something Hard(er)? Explaining Question Difficulty in Natural Language2026

Peng Cui, Qiaoyuan ZHENG (co-first author), Rudolf Debelak, Mrinmaya Sachan

Under Review at ICLR 2027

- Combined item response theory with LLM-based contrastive hypothesis generation to explain question difficulty, validating the explanations through held-out difficulty prediction and targeted question rewriting across reasoning benchmarks.

SkillEval: Learning Interpretable Ability Profiles of LLMs via Cognitive Diagnosis Models2026

Berke Arda, Peng Cui, Qiaoyuan ZHENG, Rudolf Debelak, Mubashara Akhtar, Mrinmaya Sachan

Under Review at ICLR 2027

- Proposed a framework combining LLM-based skill discovery with neural cognitive diagnosis to construct interpretable ability profiles for 3,811 LLMs across 9,523 items from five benchmarks.

Last Translation Benchmark🔗2026

Vilém Zouhar, Niyati Bafna, Mukund Choudhary, ..., Qiaoyuan Zheng, ...

Preprint

- Contributed translation test cases and reviewed English, Chinese, and French benchmark submissions.

Selected Projects

Sorbonne University

APERIODIC TILING GENERATION

Mar 2024 - Jul 2024

- Implemented Penrose, Wang, and Hat tiling algorithms in Python.
- Optimized generation through duplicate detection and spatial pruning; analyzed runtime scaling.

PythonAlgorithms

ESIEE Paris

Q-LEARNING AND SARSA IN GRID WORLDS

Jul 2023 - Aug 2023

- Implemented and compared Q-learning and SARSA across 15 custom grid-world environments.
- Built a map editor and environments adopted for master's-level teaching at UPEC.

PythonReinforcement Learning

4EU+ Alliance Research Program

COVID-19 MOBILITY ANALYSIS

Feb 2020 - Dec 2021

- Led a four-person team studying cross-country mobility patterns during COVID-19.
- Used linear regression and visualization to examine associations between public-health policies and time spent at home.

PythonStatistical Analysis

Skills

Languages

●●●●○Chinese (Native) | English (Fluent, IELTS 7.5, GRE 328) | French (Fluent, Delf B2)

Programming

●●●●●Python | C | Java | R | Ocaml | SQL

Tech Stack

Docker | Git

Frameworks & Lib

Pandas | NumPy | Scikit-learn | Pytorch

Personal Interests

Musical instruments (Drums, Saxophone, etc.) | Cooking | Reading | Calligraphy